

Xiaolong Cheng

Postdoctoral Applicant | SLAM, Scene Understanding, Multi-Robot System, Mobile Manipulation

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RESEARCH PROFILE

I am a Ph.D. candidate in Mechanical Engineering at Southeast University, advised by Prof. Keke Geng. My research focuses on robust robotic perception, including LiDAR-based SLAM, continuous-time multi-sensor fusion, 3D reconstruction, and air-ground collaboration. My doctoral work addresses sensor diversity, rapid motion, geometric and visual degeneracy, and limited cross-platform overlap. I combine algorithm development with hands-on robotics, having built a UAV-UGV platform and collected a multimodal dataset for real-world evaluation. Building on this foundation, I am interested in extending geometric perception toward spatial understanding and perception-driven robot interaction in complex environments. I am seeking a postdoctoral position in SLAM, mobile manipulation, or collaborative robotics.

RESEARCH INTERESTS

- Multi-Sensor fusion SLAM and robust localization in challenging environment
- Multi-robot exploration, coordination, and cooperative mapping
- Scene understanding, mobile manipulation, and robot-environment interaction

EDUCATION

Ph.D. Candidate in Mechanical Engineering, Southeast University

09/2021 – 01/2027

Integrated Master-PhD Program, Nanjing, China

National Scholarship (top 3%)

B.Eng. in Mechanical Engineering, Harbin University of Science and Technology

09/2017 – 06/2021

Harbin, China

Corporate Scholarship (top 1%) | Outstanding Graduate Student

SELECTED PUBLICATIONS

● Journal Articles

- [1] **X. Cheng** et al., "Onion-LO: Why Does LiDAR Odometry Fail Across Different LiDAR Types and Scenarios?," IEEE Robotics and Automation Letters (**RA-L**), 2025. Code: <https://github.com/huashu996/Onion-LO>. (**KITTI Odometry Rank: 26th**)
- [2] **X. Cheng** et al., "LIVOX-CAM: Adaptive Coarse-to-Fine Visual-Assisted LiDAR Odometry for Solid-State LiDAR," IEEE Robotics and Automation Letters (**RA-L**), 2025. Code: <https://github.com/huashu996/LIVOX-CAM>.
- [3] **X. Cheng**, K. Geng, C. Ma, Z. Lin, Y. Sun, and T. Ma*, "NDT-CAM: A Simple, Adaptive, and Robust Visual-Assisted LiDAR Odometry for High-Altitude Mapping," Robotics and Autonomous Systems (**RAS**), 2026 accepted. Code: <https://github.com/huashu996/NDT-CAM>.
- [4] **X. Cheng** et al., "SLBAF-Net: Super-Lightweight Bimodal Adaptive Fusion Network for UAV Detection in Low-Recognition Environment," Multimedia Tools and Applications, 2023. Code: <https://github.com/huashu996/SLBAF-Net>.

● Conference Papers

- [5] **X. Cheng**, Y. Sun, K. Geng, T. Ma, and Z. Liu, "Onion-LO++: An Adaptive and Degradation-Resistant Continuous-Time LiDAR Odometry," in Proceedings of the IEEE International Conference on Robotics and Automation (**ICRA**), 2026. Code: <https://github.com/huashu996/Onion-LO++>.
- [6] **X. Cheng** et al., "SVM-LO: An Accurate, Robust, Real-Time LiDAR Odometry With Segmentation Voxel Map for Autonomous Vehicles," in 2024 IEEE 27th International Conference on Intelligent Transportation Systems (**ITSC**), 2024.
- [7] **X. Cheng** et al., "Semantic Mapping Optimization Based on LiDAR and Camera Data Fusion for Autonomous Vehicle," in 2022 6th CAA International Conference on Vehicular Control and Intelligence (**CVCI**), 2022. Code: https://github.com/huashu996/camera_lidar_semantic_slam.

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ACADEMIC SERVICE

Reviewer for IEEE Robotics and Automation Letters (RA-L), IEEE Transactions on Intelligent Vehicles (T-IV)

Reviewer for ICRA, IROS, and ITSC

SELECTED PROJECTS

1. Air–Ground Collaborative SLAM | Project Leader

11/2025 – present

Developed a UAV–UGV collaborative robotic system for cooperative localization and true-color 3D reconstruction, integrating multi-sensor platform calibration, wireless inter-robot communication, and real-time collaborative SLAM for mapping, localization, and autonomous obstacle avoidance.

2. LiDAR–Vision Fusion-Based Intelligent Road Inspection System | Project Leader

03/2025 – 11/2025

Developed a public bus multi-sensor mapping system for urban elevated roads, integrating GPS, LiDAR, and vision sensors to support road surface reconstruction, road distress localization, dynamic object removal, and quantitative distress area estimation through LiDAR–camera SLAM and visual detection fusion, resulting in **publication [2]**.

3. AVP-SLAM Localization System Development | Project Leader

10/2024 – 02/2025

Led the development of a LiDAR-based autonomous valet parking system using a pre-built global map of a multi-level underground parking garage. The project involved constructing a high-resolution 3D point cloud map, developing a pure LiDAR relocalization algorithm for AVP scenarios, and designing an adaptive LiDAR odometry method to improve robustness across different parking environments, resulting in **publication [2]**.

4. Mining Area Volume Estimation Based on UAV LiDAR SLAM | Project Leader

02/2024 – 06/2024

Led the development of a UAV-based LiDAR–camera data acquisition and mapping system for mining-area reconstruction and stockpile volume estimation. The project involved integrating LiDAR and visible-light cameras on a UAV platform, designing a LiDAR–camera–IMU fusion mapping algorithm for accurate 3D reconstruction, and performing point cloud densification, region-of-interest cropping, and volumetric analysis, resulting in **publication [3]**.

5. Intelligent Vehicle Perception in Low-Visibility Conditions

12/2021 – 12/2022

Led the development of a visible–infrared bimodal perception system for long-range traffic target detection, covering vehicles and pedestrians within 300 m. The project involved building the bimodal sensing platform, collecting and annotating a visible–infrared traffic object detection dataset, and designing the Dual-YOLO bimodal detection network, , resulting in **publication [4]**.

TECHNICAL EXPERTISE

- **Programming:** C++, Python, PyTorch, ROS/ROS2, Linux, Docker
- **Algorithmic Skills:** Proficient in applying LO/LIO/LIVO frameworks for point cloud registration, state estimation, loop closure, and back-end optimization, and in neural scene reconstruction using 3D Gaussian Splatting. Experienced in deploying SLAM systems on real robots, integrating visual perception (YOLO, DeepLabV3+, YOLACT) for scene understanding, and supporting downstream planning and mobile manipulation.
- **Libraries & Tools:** PCL, OpenCV, Eigen, Sophus, Ceres Solver, GTSAM, RViz, Gazebo, OpenMP, TBB
- **Platforms & Sensors:** UAVs, UGVs, LiDAR-Camera-IMU Systems, GPS/RTK, Infrared Cameras, Livox, Hesai, Ouster, RoboSense

SELECTED HONORS

National Gold Award, Intelligent Driving Challenge, 2022

National Second Prize, 19th Huawei Cup Postgraduate Mathematical Contest in Modeling, 2022

International Meritorious Winner, Mathematical Contest in Modeling (MCM), 2020

National Second Prize, National Undergraduate Mathematical Contest in Modeling, 2019

REFERENCES

Keke Geng, Associate Professor, Southeast University | jsgengke@seu.edu.cn

Guodong Yin, Professor, Southeast University | ygd@seu.edu.cn

Weichao Zhuang, Professor, Southeast University | wezhuang@seu.edu.cn

Multi-agent collaborative perception, localization and mapping

Reliable localization and consistent maps for heterogeneous robots under challenging sensing and motion conditions.

RESEARCH PROGRESSION (LO → LIO → LIVO → Collaborative SLAM)

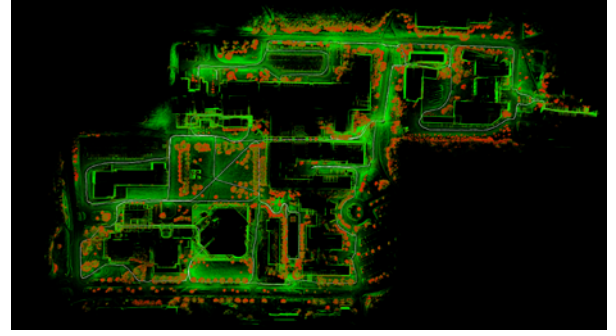
01

Sensor-Adaptive LiDAR Odometry

Problem: LiDAR odometry exhibits poor generalizability and limited adaptability across different LiDAR sensors and environments.

Method: Unified adaptive point cloud preprocessing structure.

Outcome: LiDAR-only odometry adapts to diverse environments and LiDAR sensors with different scanning patterns without parameter tuning.



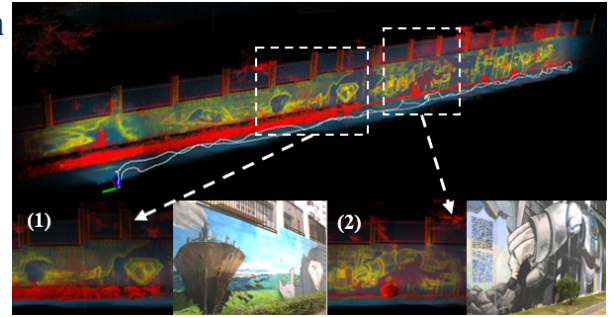
02

Odometry for high-speed and aggressive motion

Problem: Rapid motion and within-scan distortion.

Method: Continuous-time trajectory and dynamic IMU initialization.

Outcome: Adapts to high-speed, high-frequency motion and supports dynamic IMU initialization.



03

Tightly coupled multi-sensor odometry

Problem: Degenerate environments, real-time performance, and multi-sensor data synchronization.

Method: Unified 21D continuous-time trajectory optimization, visual anchor extraction and tracking, and degeneracy detection.

Outcome: Robust localization and color reconstruction in complex environments.



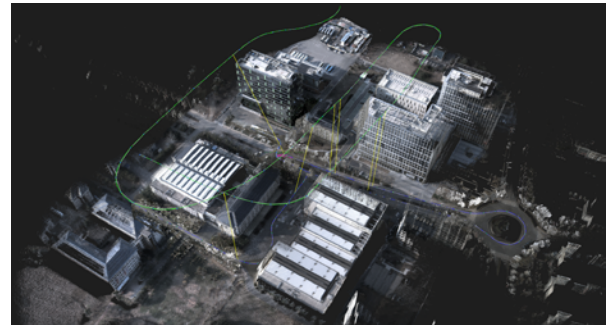
04

Air–Ground Collaborative SLAM

Problem: Large viewpoint disparities, map ghosting in collaborative mapping, and poor global map consistency.

Method: Cross-robot collaborative factors, loop closure detection, and joint global optimization.

Outcome: Generates globally consistent color maps in complex, heterogeneous aerial–ground collaborative environments.



Self-developed UAV–UGV experimental platform and dataset

